# ChE 2O04 Tutorial 3

**Tutorial Problem #1**

A 10 cm diameter steel ball is suspended by a string in a vessel filled with two liquids, water and carbon tetrachloride. The fluids are immiscible and form two layers: the bottom layer is 30 cm deep and the top layer is 35 cm deep.

a) Draw a diagram of the process with the two layers labelled as the appropriate fluids.

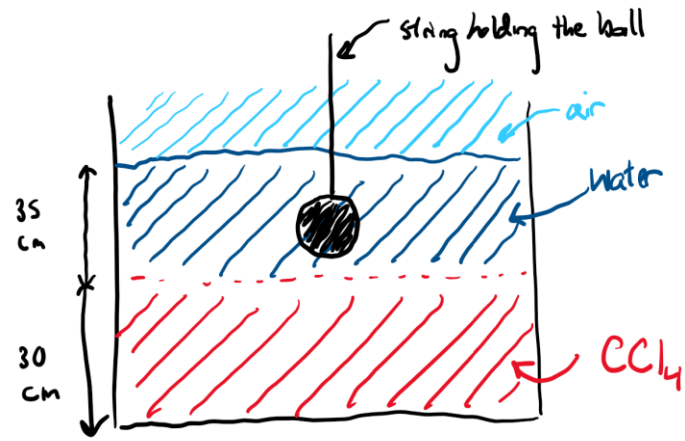



*Table A.1*
**Specific Gravities of Selected Engineering Materials** *(continued)*

(b) Common Materials

| Material | Specific Gravity (—) |
|---|---|
| Aluminum | 2.64 |
| Balsa wood | 0.14 |
| Brass | 8.55 |
| Cast Iron | 7.08 |
| Concrete (cured) | 2.4[a] |
| Concrete (liquid) | 2.5[a] |
| Copper | 8.91 |
| Ice (0°C) | 0.917 |
| Lead | 11.4 |
| Oak | 0.77 |
| Steel | 7.83 |
| Styrofoam (1 pcf[b]) | 0.0160 |
| Styrofoam (3 pcf) | 0.0481 |
| Uranium (depleted) | 18.7 |
| White pine | 0.43 |

*Source:* Data from Reference [4].
[a]depending on aggregate.
[b]pounds per cubic foot.

*Table A.2*
**Physical Properties of Common Liquids at 20°C**

| Liquid | Isentropic Bulk Modulus[a] ($GN/m^2$) | Specific Gravity (—) |
|---|---|---|
| Benzene | 1.48 | 0.879 |
| Carbon tetrachloride | 1.36 | 1.595 |
| Castor oil | 2.11 | 0.969 |
| Crude oil | — | 0.82–0.92 |
| Ethanol | — | 0.789 |
| Gasoline | — | 0.72 |
| Glycerin | 4.59 | 1.26 |
| Heptane | 0.886 | 0.684 |
| Kerosene | 1.43 | 0.82 |
| Lubricating oil | 1.44 | 0.88 |
| Methanol | — | 0.796 |
| Mercury | 28.5 | 13.55 |
| Octane | 0.963 | 0.702 |
| Seawater[b] | 2.42 | 1.025 |
| SAE 10W oil | — | 0.92 |
| Water | 2.24 | 0.998 |

*Source:* Data from References [1, 5, 6].
[a]Calculated from speed of sound; 1 $GN/m^2 = 10^9$ $N/m^2$ (1 $N/m^2 = 1.45 \times 10^{-4}$ $lbf/in.^2$).
[b]Dynamic viscosity of seawater at 20°C is $\mu = 1.08 \times 10^{-3}$ $N \cdot s/m^2$. (Thus, the kinematic viscosity of seawater is about 5 percent higher than that of freshwater.)

Given that the specific gravity of carbon tetrachloride is greater than water's, the carbon tetrachloride would sink and form a layer beneath the water layer.

b) Draw a force diagram on the ball (hint: there are three forces due to gravity, buoyancy, and tension)

Where $F_B$ is the force of buoyancy, $F_g$ is the force of gravity, and $F_T$ is the tension force.

c) Calculate the tension in the string when the ball is completely submerged in carbon tetrachloride.
d) Calculate the tension in the string when the ball is submerged half in water and half in carbon tetrachloride.
e) Calculate the tension in the string when the ball is submerged half in the top layer and half in air (i.e., the gas above the top layer).

For all three cases above, the following force balance applies:

$$F_T + F_B = F_g$$

That means, to determine the tension force in each case:

$$F_T = F_g - F_B$$

$F_g$ is the same in all cases, as determined by:

$$F_g = \rho \times g \times V,$$

where $\rho$ is the density of the steel [7830 $kg/m^3$], $g$ is the gravitational constant [9.81 $m/s^2$], and volume can be calculated as the volume of a sphere based on the diameter.

$F_B$ can be calculated for each case through the following equation:

$$F_B = \rho_{fluid} \times g \times V_{object\ submerged}$$

In the case where the ball is partially submerged, the $F_B$ can be calculated as:

$$F_B = (\rho_{CCL4} \times g \times \frac{1}{2} V_{object\ submerged}) + \rho_{water} \times g \times \frac{1}{2} V_{object\ submerged}$$

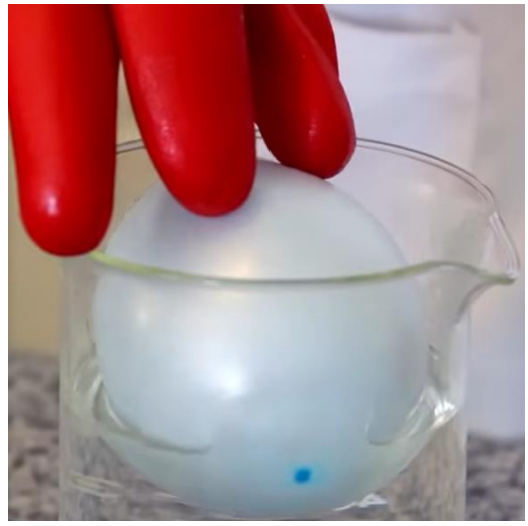
The same equation applies for when the need is half submerged in water and air, with the modified density values.

| | | SUBMERGED IN CCl4 | | | | SUBMERGED HALF WATER-HALF CCl4 | | | | SUBMERGED HALF WATER-HALF AIR | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| constant | | | 9.81 | m/s2 | | | | 9.81 | m/s2 | | | 9.81 | m/s2 |
| | | | 1.21 | kg/m3 | | | | 1.21 | kg/m3 | | | 1.21 | kg/m3 |
| | top | water | 998 | kg/m3 | | top | water | 998 | kg/m3 | top | water | 998 | kg/m3 |
| | bottom | ccl4 | 1595 | kg/m3 | | bottom | ccl4 | 1595 | kg/m3 | bottom | ccl4 | 1595 | kg/m3 |
| | | steel | 7830 | kg/m3 | | | steel | 7830 | kg/m3 | | steel | 7830 | kg/m3 |
| | | diameter | 0.1 | m | | | diameter | 0.1 | m | | diameter | 0.1 | m |
| | | Volume | 0.000524 | m3 | | | Volume | 0.000524 | m3 | | Volume | 0.000524 | m3 |
| | | Gravity | 40.2 | N | | | Gravity | 40.2 | N | | Gravity | 40.2 | N |
| | | Buoyancy | 8.2 | N | | | Buoyancy | 6.7 | N | | Buoyancy | 2.6 | N |
| | | Tension | 32.0 | N | | | Tension | 33.6 | N | | Tension | 37.7 | N |

The resulting calculations demonstrate that in comparison to the case where the full ball is submerged in tetrachloride, as the ball is in less dense liquids, the magnitude of the buoyancy force decreases, resulting in a greater tension force required to suspend the ball. Similarly, as the ball moves towards into the denser liquid, the impact of buoyancy increases, reducing the required tension force.

**Tutorial Problem #2**

The pressure of a gas is a function of its density and temperature, as described by the ideal gas law (Eq. 1.1 in the textbook). A good example of this is a balloon exposed to hot and cold temperatures, as shown in the attached video: (https://www.youtube.com/watch?v=NplVuTrr59U&t=158s)



In lecture it was shown that for the special case of isothermal (constant temperature), the resulting equation for pressure variation was:

$$P_2 = P_1 \exp\left[-\frac{gM_w}{R_U T}(z_2 - z_1)\right]$$

**Part A:** Following a similar approach to that presented in lecture for the isothermal case, derive the general expression for pressure as a function of elevation for the general case of linear temperature variation:

Up to 11.0 km in the U.S. Standard Atmosphere, temperature decreases linearly with the following relationship:

$$T = mz + T_o \text{ where } T_o \text{ is the temperature at sea level}$$

From equation 3.6 in the textbook, we have the Ideal Gas Law:

$$\frac{dP}{dz} = -\rho g$$

$$\frac{dP}{dz} = -(\frac{M_w P}{R_u T})g$$

By implementing separation of variables:

$$\frac{1}{P}dP = -(\frac{M_w g}{R_u}) \times \frac{1}{T}dz$$

Subbing in the temperature decrease with elevation:

$$(\frac{1}{P})dP = -(\frac{M_w g}{R_u})(\frac{1}{mz + T_o})dz$$

By subbing in our integration limits:

$$\int_{P1}^{P2} (\frac{1}{P})dP = -(\frac{M_w g}{R_u}) \int_{z1}^{z2} (\frac{1}{mz + T_o})dz$$

Taking the integral of each side:

$$\ln(P_2) - \ln(P_1) = -\left(\frac{M_w g}{R_u}\right) \times \frac{1}{m} \times [\ln(mz_2 + T_o) - \ln(mz_1 + T_o)]$$

Using the quotient properties of logarithms:

$$\ln\left(\frac{P_2}{P_1}\right) = -\left(\frac{M_w g}{mR_u}\right) \times \ln\left(\frac{mz_2 + T_o}{mz_1 + T_o}\right)$$

Taking the exponent of both sides:

$$\frac{P_2}{P_1} = \left[\frac{mz_2 + T_o}{mz_1 + T_o}\right]^{-\left(\frac{M_w g}{mR_u}\right)}$$

**Part B:** The special case of linear temperature variation with height is exactly what happens in the troposphere, the lowest level of Earth's atmosphere. Using the data in the table below, calculate the value of the constant $m$ in the equation:

$$T = mz + T_o$$

| Elevation (metres) | Temperature (Kelvin) |
|---|---|
| 0 | 288.15 |
| 11,000 | 216.65 |

*Hints: The constant m in the equation above is the best-fit value of the slope of temperature versus elevation.*
*$T_o$ is the temperature at sea level (i.e. z = 0)*

$$T = mz + T_o$$

Given that $T_o$ = 288.15k and $T$ = 216.65k, and that $z = 11{,}000$ m:

$$\frac{T - T_0}{z} = m$$

$$\frac{216.65 - 288.15}{11{,}000} = m = -0.0065 \text{ K/m.}$$

**Part C:** Now substitute the value for $m$ found in Part B into the formula derived in Part A and use the resulting expression to make a plot of atmospheric pressure versus elevation.

- Set the x-axis range from 0 to 9,000 m
- Set the y-axis range from 0 to 120 kPa

*Hint: Use the following values in your calculation:*

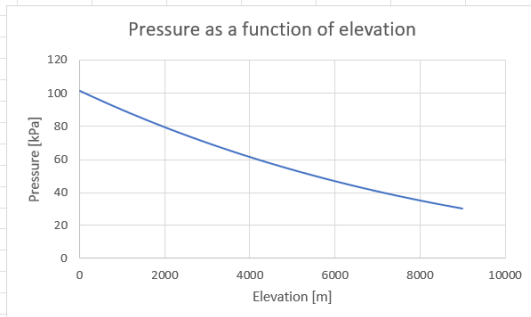
$M_w$ = 28.97 kg/kg-mol
$R_u$ = 8314 J/kg-mol/K

Using the equation developed in Part A, and subbing in the correct constants (g = 9.81 m/$s^2$ and $M_w$ = 28.97 kg/kg-mol and $R_u$ = 8314 J/kg-mol/K; $m$ = -0.0065 K/m; $T_0$ = 288.15 K and $Z_1 = 0$:

$$P_2[kPa] = 101.3 \times [\frac{(-0.0065z_2 + 288.15}{288.15}]^{-\left(\frac{(28.97)*(9.81)}{(-0.0065)*(8314)}\right)}$$

The expression above can be inputted into excel [as above] to then create the below graph:

| | | | | |
|---|---|---|---|---|
| | | g | 9.81 | m/s^2 |
| | | MW | 28.97 | g/mol |
| | | m | -0.0065 | K/m |
| | | RU | 8314 | J/kg-mol*k |
| | | T1 | 288.15 | K |
| | Elevation (m) | Pressure [kPa] | | |
| Varying T | 0 | 101.325 | | |
| | 100 | 100.1288 | | |
| | 200 | 98.94398 | | |
| | 300 | 97.77058 | | |
| | 400 | 96.60847 | | |
| | 500 | 95.45758 | | |
| | 600 | 94.31781 | | |
| | 700 | 93.18909 | | |
| | 800 | 92.07133 | | |
| | 900 | 90.96446 | | |
| | 1000 | 89.86839 | | |
| | 1100 | 88.78304 | | |
| | 1200 | 87.70833 | | |




**Part D:** Justify and check the plot in part D through the following statement: "the atmospheric pressure at the summit of Everest is roughly 1/3 of that at sea level".

If $P_1$ at z = 0, and $P_2$ at z = 9000 m are compared, the ratio of $P_2$ to $P_1$ is 0.3032, meaning that the atmospheric pressure at 9000m is 30.32% of the pressure at sea level, matching the statement in Part D.

The pressure at $P_1$ is 101.3 kPa and the pressure at $P_2$ is 30.7 kPa.